

Roll No. 226010256021

71478

**B. Pharma. 7th Semester
(Regular/Re-Appear/Mercy Chance)
Examination – December, 2025**

NOVEL DRUG DELIVERY SYSTEM

Paper : BP704T

Time : Three Hours]

[Maximum Marks : 75

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is *compulsory*, each carry 2 marks, Attempt any *seven* questions out of nine from Section-B, each carry 5 marks and *two* questions out of three from Section-C, each carry 10 marks.

SECTION – A

1. Attempt *all* questions.

10 × 2 = 20

(a) Write a note on permeation enhancer.

(b) Explain the advantage of natural polymers.

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- (c) What is microencapsulation ?
- (d) Write a note on microspheres.
- (e) What do you mean by NDDS ?
- (f) Write the merits and demerits of mucosal drug delivery system.
- (g) Discuss about osmotic pump.
- (h) Write a note on diffusion.
- (i) Write a note on metered dose.
- (j) Discuss biological properties of drug relevant to CDDS.

SECTION – B

Note : Answer any *seven* out of the following nine questions :

$$7 \times 5 = 35$$

- 2. What is the factor affecting permeation in transdermal drug delivery system ?
- 3. Discuss in detail about nebulizer.
- 4. Write a note on intraocular barriers.
- 5. Explain in detail about liposomes.
- 6. Discuss the method of microencapsulation and its application.

- ✓ 7. Discuss about implantable drug delivery system.
- ✓ 8. Explain the classification and properties of polymers.
- ✓ 9. Write a note on control drug delivery system.
10. Discuss the principle of bioadhesion and mucoadhesion.

SECTION – C

Note : Answer any *two* out of the following three questions : $2 \times 10 = 20$

11. Discuss in details about targeted drug delivery system.
 12. Explain gastroretentive drug delivery system in detail.
 13. What are Intrauterine drug delivery system discuss in detail ?
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DEPARTMENT OF PHARMACEUTICAL SCIENCES, INDIRA GANDHI UNIVERSITY, MEERPUR, REWARI
First Sessional Theory Examination of B. Pharmacy 7th Semester –October, 2025
Subject: Novel drug delivery system Theory (Paper Code: BP704T)

Time: 1.0 hr

Max. Marks =30

Attempt all questions

Attempt all questions					5 × 2 =10
Q1	Give the answers of all questions				
	i.	Differentiate matrix and reservoir	iv	What is the range of microcapsules? What size of particles are suitable for air suspension coating?	
	ii.	State applications of polymers in CRDDS	v	Write 2 advantages and 2 disadvantages of mucoadhesive DDS.	
iii.	Name the three immiscible phases in coacervation.				
Q2.	Attempt any two question (Short answer type questions)				2 × 5 = 10
	a)	Define polymer and classify them.			
	b)	Describe any 2 theories of mucoadhesion.			
	c)	Explain osmotically pressure activated DDS.			
Q3	Attempt any one question (Long answer type questions)				1 × 10 = 10
	a)	Explain the design involved in controlled release formulations.			
	b)	Describe in detail buccal and sublingual drug delivery with their advantages and disadvantages.			